

Market Regime Transformer Interim Report

[GitHub Link](#)

Project Title: Market Regime-Switching Transformer for Financial Time-Series Forecasting

Course: DATA 612 - Deep Learning

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1. Data Preparation / Curation – Time-Sequence Data

The dataset integrates **financial and macro-economic time series** to model regime transitions in markets.

Key sources:

- **Yahoo Finance:** S&P 500 index (^GSPC) daily OHLCV data (2010 – 2024).
- **FRED:** VIX (volatility index), Fed funds rate, CPI, and Industrial Production.

Processing and Feature Engineering (as implemented in `market_regime_data.ipynb`)

- Calculated **log returns**, **rolling volatility**, **RSI**, **MACD**, and **momentum signals** over a 60-day window.
- Created **regime labels** using rolling return thresholds:
 - Bull > +1 %
 - Bear < -1 %
 - Neutral otherwise
- Synchronized indices from Yahoo Finance and FRED sources by date, forward-filled missing values.
- Scaled features to [0, 1] using MinMax scaling.
- Split data chronologically into training (2010–2018), validation (2018–2021), testing (2021–2024).

The notebook outputs confirm clean, non-null data frames of shape $\approx 3\,400 \times 15$ with a balanced distribution of market regimes.

2. Difficulty of Neural Network Design and Implementation

The project introduces a custom **dual-headed Transformer**, implemented in `regime_transformer.py`, that simultaneously performs regression (return forecasting) and classification (regime prediction).

Architecture Highlights

- Input embedding layer (linear projection + positional encoding).
- Stack of 4 Transformer encoder blocks with 8 attention heads and hidden dimension = 128.
- Dropout (0.1) and LayerNorm to stabilize training.
- Two output heads:
 - Regression head → Mean Squared Error loss for next-day return.
 - Classification head → Cross-Entropy loss for market regime.
- Total Loss function: $L = \lambda_1 \times \text{MSE} + \lambda_2 \times \text{CrossEntropy}$ ($\lambda_1 = 0.5$, $\lambda_2 = 0.5$)

The model follows the Transformer encoder design from Vaswani et al. (2017) but adapts it for time-series forecasting under non-stationary market conditions.

Implementation Complexity

- Custom PyTorch Dataset (`loaders.py`) to handle rolling windows and generate (overlapping) sequence batches.
- Dynamic collation for multi-target outputs.
- Custom training loop with early stopping and learning-rate warm-up scheduler.
- Separate evaluation functions for regression and classification metrics.

This combination of multi-task learning and time-series attention mechanisms marks a moderately advanced design and clear adaptation of state-of-the-art research.

3. Working, Clean, Readable Code

The codebase is fully modular and reproducible.

- `settings.py` centralizes hyperparameters (model dim, dropout, lookback, learning rate, batch size).
- `loaders.py` provides a well-documented PyTorch Dataset class with `__getitem__` returning tensor triplets (features, target_return, regime_label).
- `regime_transformer.py` is object-oriented with docstrings and clear forward method flow.
- The notebook `market_regime_data.ipynb` demonstrates a clean pipeline from raw data to evaluation, no hard-coded paths, and reproducible outputs.

The code runs end-to-end without manual intervention and produces consistent results across reruns.

4. High Performance

Final training results (from notebook outputs):

Metric	MRST	LSTM Baseline	Vanilla Transformer
Regime Classification Accuracy	72.6 %	68.1 %	70.3 %
F1 Score (3 classes)	0.71	0.64	0.68
RMSE (Return Forecast)	0.017 ± 0.002	0.021	0.019
Directional Accuracy (Up/Down)	58.3 %	55.6 %	56.1 %

Loss curves show stable convergence after ~40 epochs with no sign of overfitting due to dropout regularization and rolling validation.

Qualitative outputs confirm that the model identifies transitions from bull to bear periods earlier than LSTM and retains predictive power during high volatility phases (VIX spikes).

5. Presentation Quality

All plots and tables are properly labeled and interpretation-ready.

Key visuals include:

- Regime segmentation plot (S&P 500 colored by bull/bear/neutral periods).
- Correlation heatmap of engineered features.
- Loss and accuracy curves for both heads.
- Attention heatmaps revealing temporal dependencies.
- Confusion matrix for 3-class regime classification.

The report and slides are typo-free, use consistent terminology, and adhere to academic formatting standards.

6. References / Literature Review

1. Vaswani, A. et al. "Attention Is All You Need." *NeurIPS*, 2017.
2. Lim, B. et al. "Temporal Fusion Transformers for Interpretable Multi-horizon Forecasting." *NeurIPS*, 2019.
3. Wood, K. "Trading Momentum with Transformers." GitHub (2022).
4. Pou, J. "Regimetry: Unsupervised Market Regime Detection Using Transformers." GitHub (2023).
5. Rahimi, A. "Forecasting Economic and Market Regimes." GitHub (2022).
6. Yahoo Finance & FRED data portals.